1 Erdős problem 477

1.1 Introduction

A set $A \subset \mathbb{Z}$ is a *tiling complement* for a set $B \subset \mathbb{Z}$ if every integer n has a unique representation $n = a + b$ with $a \in A$ and $b \in B$. Erdős and Graham asked whether there is a polynomial $f : \mathbb{Z} \to \mathbb{Z}$ of degree at least two for which the value set $\{f(k) : k \in \mathbb{Z}\}$ has a tiling complement; see [EG80, p. 95] and also the catalogue entry [Bl]. We prove the following result.

Theorem 1.1. *Let*

$$B = \{m^{13} : m \in \mathbb{Z}\}.$$

There exists a set $A \subset \mathbb{Z}$ such that every $n \in \mathbb{Z}$ is represented uniquely in the form

$$n = a + m^{13}, \qquad a \in A, \quad m \in \mathbb{Z}.$$

Equivalently, the set of thirteenth powers has a tiling complement in $\mathbb{Z}$.

We write

$$D = B - B = \{u^{13} - v^{13} : u, v \in \mathbb{Z}\}.$$

The proof has three components. First, a Brownawell–Masser S-unit estimate over the rational function field rules out nonconstant rational curves on the diagonal surface

$$u^{13} - v^{13} - t^{13} = -c$$

whenever $c \notin B$. Second, Heath-Brown's determinant-method estimate then gives a power-saving bound for the number of shifts t^{13} for which $t^{13} - c \in D$. Finally, this power saving verifies a simple greedy criterion for constructing a tiling complement.

Throughout, the notation $U \ll_c V$ means that $|U| \leq C(c)V$ for a constant depending at most on c; similarly for $O_c(V)$.

1.2 External inputs

We use two standard estimates. The first is the Brownawell–Masser theorem for S-unit equations over function fields, specialized to genus zero.

Theorem 1.2 (Brownawell–Masser on $\mathbb{P}^1$)**.** *Let k be an algebraically closed field of characteristic zero, and let S be a finite set of points of $\mathbb{P}^1_k$. Let $u_1, \ldots, u_r \in k(t)^\times$ be S-units, with $r \geq 3$, not all constant, satisfying*

$$u_1 + \cdots + u_r = 0,$$

and suppose that no proper nonempty sub-sum vanishes. With the projective height convention

$$H(u_1 : \cdots : u_r) = -\sum_{P \in \mathbb{P}^1_k} \min_{1 \leq i \leq r} \operatorname{ord}_P(u_i),$$

one has

$$H(u_1 : \cdots : u_r) \le \binom{r-1}{2}(|S|-2).$$

For $r = 3$ this is the Mason–Stothers inequality, with coefficient 1.

This is the genus-zero case of [BM86]; the height convention and the three- and four-term constants are recalled explicitly in [CZ11].

The second input is the form of Heath-Brown's theorem needed below.

Theorem 1.3 (Heath-Brown)**.** *Let $F \in \mathbb{Z}[X_1, X_2, X_3]$ be a nonsingular ternary form of degree $k \ge 3$. Let $X \ge 1$, and let N be a fixed nonzero integer with $|N| \ll_F X$. The number of integer solutions of*

$$F(x_1, x_2, x_3) = N, \qquad \max_i |x_i| \le X,$$

which do not lie on a nonconstant polynomial parametrization of degree at most $\lfloor k/10 \rfloor$ is

$$O_F(X^{10/k}).$$

Here a polynomial parametrization means a triple of polynomials $p_1, p_2, p_3 \in \mathbb{Z}[T]$, not all constant, such that $F(p_1(T), p_2(T), p_3(T)) = N$.

This is Theorem 2 of [HB09]; for diagonal forms Heath-Brown also formulates the result as an estimate apart from the "special" solutions.

1.3 A function-field exclusion of rational curves

Lemma 1.4. *Let $N \in \mathbb{Q}^\times$, and consider the diagonal surface*

$$X_N : \quad X^{13} + Y^{13} + Z^{13} = NW^{13} \subset \mathbb{P}^3_{\mathbb{Q}}.$$

If $N \notin \mathbb{Q}^{13}$, then X_N contains no nonconstant $\mathbb{Q}$-rational parametrized curve. If $N = d^{13}$ with $d \in \mathbb{Q}^\times$, then the images of nonconstant morphisms $\mathbb{P}^1_{\mathbb{Q}} \to X_N$ are precisely the three lines

$$X = -Y, Z = dW; \qquad X = -Z, Y = dW; \qquad Y = -Z, X = dW.$$

Proof. Suppose that a nonconstant morphism $\phi : \mathbb{P}^1_{\mathbb{Q}} \to X_N$ exists. Choose homogeneous coordinates $[S : T]$ on $\mathbb{P}^1$ and write

$$\phi = [F : G : H : R],$$

where $F, G, H, R \in \mathbb{Q}[S, T]$ are homogeneous forms of the same degree $e \ge 1$, with no common zero. Zero coordinate forms are allowed. The defining equation gives

$$F^{13} + G^{13} + H^{13} - NR^{13} = 0. \tag{1.1}$$

We apply Theorem 1.2 after extending scalars to $\overline{\mathbb{Q}}$.

First assume that all four terms in (1.1) are nonzero and that no proper nonempty sub-sum vanishes. Divide by F^{13}. The four functions

$$1, \qquad (G/F)^{13}, \qquad (H/F)^{13}, \qquad -N(R/F)^{13}$$

are S-units for the set S of zeros of $FGHR$ on $\mathbb{P}^1_{\overline{\mathbb{Q}}}$. Since each of F, G, H, R has degree e, one has $|S| \le 4e$. Moreover, because the four coordinate forms have no common zero,

$$H(F^{13} : G^{13} : H^{13} : R^{13}) = 13e.$$

Multiplication of a coordinate by the nonzero constant N does not change this height. The four-term case of Theorem 1.2 therefore gives

$$13e \le 3(|S| - 2) \le 3(4e - 2) = 12e - 6,$$

which is impossible.

Now suppose exactly three terms are nonzero and no proper sub-sum vanishes. Let S be the set of zeros of the product of the corresponding three coordinate forms. Then $|S| \le 3e$, the height is $13e$, and the three-term Mason–Stothers case of Theorem 1.2 gives

$$13e \le |S| - 2 \le 3e - 2,$$

again impossible. If exactly three nonzero terms have a proper vanishing sub-sum, then a minimal such sub-sum has length two, and the complementary one-term sub-sum would also vanish, contradicting the assumption that the remaining term is nonzero. Thus exactly three nonzero terms cannot occur.

If at most two terms are nonzero, equation (1.1) forces the ratio of the two nonzero coordinate forms to be constant; hence the image of ϕ is a point. This is incompatible with nonconstancy. Consequently, for a nonconstant morphism all four terms are nonzero and some proper nonempty sub-sum vanishes.

A proper vanishing sub-sum cannot have length one. It also cannot have length three, because its complementary one-term sub-sum would then vanish. Hence the four terms split into two vanishing pairs. For example, the pairing

$$F^{13} + G^{13} = 0, \qquad H^{13} - NR^{13} = 0$$

implies $F = -G$, since $\mathbb{Q}$ contains no nontrivial thirteenth root of unity and 13 is odd. The second equality gives $(H/R)^{13} = N$ in $\mathbb{Q}(S/T)$; hence H/R is constant and $N = d^{13}$ for some $d \in \mathbb{Q}^\times$. To make this point explicit, view H/R as a rational function on $\mathbb{P}^1_{\overline{\mathbb{Q}}}$. For a point $P \in \mathbb{P}^1_{\overline{\mathbb{Q}}}$, let $\mathrm{ord}_P(f)$ denote the order of a nonzero rational function f at P: it is positive if f has a zero at P, negative if f has a pole at P, and zero otherwise. Since $H^{13} = NR^{13}$, we have $\left(\frac{H}{R}\right)^{13} = N$ in $\mathbb{Q}(S/T)$. Taking orders at any $P \in \mathbb{P}^1_{\overline{\mathbb{Q}}}$, we obtain

$$13\,\mathrm{ord}_P\left(\frac{H}{R}\right) = \mathrm{ord}_P(N) = 0. \tag{1.2}$$

Indeed, $N \in \mathbb{Q}^\times$ is a nonzero constant, so it has no zeros or poles on $\mathbb{P}^1_{\overline{\mathbb{Q}}}$. Therefore

$$\mathrm{ord}_P\left(\frac{H}{R}\right) = 0$$

for every P. Hence H/R has neither zeros nor poles on $\mathbb{P}^1_{\overline{\mathbb{Q}}}$, so H/R is constant. Since $H/R \in \mathbb{Q}(S/T)$, this constant lies in $\mathbb{Q}$. Thus

$$\frac{H}{R} = d \qquad \text{for some } d \in \mathbb{Q}^\times,$$

and consequently

$$N = d^{13}.$$

The image is then the line $X = -Y$, $Z = dW$. The other pairings give the lines $X = -Z$, $Y = dW$ and $Y = -Z$, $X = dW$ in the same way.

Thus a nonconstant $\mathbb{Q}$-rational parametrized curve forces $N \in \mathbb{Q}^{13}$, and in that case its image is one of the three displayed lines. Conversely, if $N = d^{13}$, the displayed lines are plainly contained in X_N and are $\mathbb{Q}$-rational. This proves the lemma. $\square$

Corollary 1.5. *Let $c \in \mathbb{Z} \setminus B$. The affine surface*

$$u^{13} - v^{13} - t^{13} = -c$$

has no nonconstant rational parametrization over $\mathbb{Q}$.

Proof. After the substitution $X = u$, $Y = -v$, $Z = -t$, its projective closure is

$$X^{13} + Y^{13} + Z^{13} = (-c)W^{13}.$$

Since 13 is odd, $-c \in \mathbb{Q}^{13}$ if and only if $c \in \mathbb{Q}^{13}$. For an integer c, membership in $\mathbb{Q}^{13}$ is equivalent to $c = m^{13}$ for some $m \in \mathbb{Z}$. Thus $c \notin B$ implies $-c \notin \mathbb{Q}^{13}$, and Lemma 1.4 applies. $\square$

1.4 The bad-shift estimate

For $c \in \mathbb{Z} \setminus B$ and $T \geq 1$, define

$$S_c(T) = \{t \in \mathbb{Z} : |t| \leq T,\ t^{13} - c \in B - B\}.$$

Proposition 1.6. *For every $c \in \mathbb{Z} \setminus B$,*

$$|S_c(T)| = O_c(T^{5/6}) = o(T).$$

Proof. If $t \in S_c(T)$, then for some $u, v \in \mathbb{Z}$,

$$t^{13} - c = u^{13} - v^{13}, \qquad \text{or equivalently} \qquad u^{13} - v^{13} - t^{13} = -c. \tag{1.3}$$

Since $c \notin B$, we cannot have $u = v$, for then $t^{13} = c$.

Write

$$u^{13} - v^{13} = (u - v)Q(u, v), \qquad Q(u, v) = u^{12} + u^{11}v + \cdots + uv^{11} + v^{12}.$$

For real $(u, v) \neq (0, 0)$, one has $Q(u, v) > 0$: it is the difference quotient of the strictly increasing function $x \mapsto x^{13}$, with the limiting value $13u^{12}$ on the diagonal $u = v$. By compactness on the set $\max(|u|, |v|) = 1$, there is a constant $\kappa > 0$ such that

$$Q(u, v) \geq \kappa \max(|u|, |v|)^{12} \qquad ((u, v) \in \mathbb{R}^2).$$

As $u \neq v$, we have $|u - v| \geq 1$, and hence

$$\kappa \max(|u|, |v|)^{12} \leq |u^{13} - v^{13}| = |t^{13} - c| \ll_c T^{13}.$$

Thus

$$\max(|u|, |v|) \ll_c T^{13/12}. \tag{1.4}$$

Set

$$x = u, \qquad y = -v, \qquad z = -t.$$

Then (1.3) becomes

$$x^{13} + y^{13} + z^{13} = -c. \tag{1.5}$$

By (1.4) and $|t| \le T$, every solution obtained from an element of $S_c(T)$ satisfies

$$\max(|x|, |y|, |z|) \le X, \qquad X = C_c T^{13/12}$$

for a suitable constant C_c.

Let $\varepsilon_c = \operatorname{sgn}(-c) \in \{\pm 1\}$. Equation (1.5) is equivalent to

$$\varepsilon_c(x^{13} + y^{13} + z^{13}) = |c|.$$

The form $\varepsilon_c(X_1^{13} + X_2^{13} + X_3^{13})$ is nonsingular of degree 13. For all sufficiently large X one has $|c| \ll X$; the remaining bounded range of X is absorbed by the implied constant. Since $\lfloor 13/10 \rfloor = 1$, Theorem 1.3 excludes only nonconstant polynomial parametrizations of degree at most one. Corollary 1.5 excludes nonconstant rational parametrizations of any degree on (1.5). Hence no solutions of (1.5) are lost in applying Theorem 1.3, and we obtain

$$\#\{(x, y, z) \in \mathbb{Z}^3 : x^{13} + y^{13} + z^{13} = -c, \max(|x|, |y|, |z|) \le X\} \ll_c X^{10/13}.$$

Each $t \in S_c(T)$ gives at least one such triple (x, y, z), so

$$|S_c(T)| \ll_c X^{10/13} \ll_c \left(T^{13/12}\right)^{10/13} = T^{5/6}.$$

This proves the proposition. $\square$

1.5 A greedy tiling criterion

Lemma 1.7. *Let $B \subset \mathbb{Z}$ and $D = B - B$. Suppose that for every finite set $C \subset \mathbb{Z} \setminus B$ there exists $b \in B$ such that*

$$(C - b) \cap D = \varnothing. \tag{1.6}$$

Then B has a tiling complement in $\mathbb{Z}$.

Proof. Enumerate $\mathbb{Z}$ as $n_1, n_2, \ldots$. We inductively construct finite sets

$$A_0 \subset A_1 \subset A_2 \subset \cdots$$

such that the translates $a + B$ with $a \in A_j$ are pairwise disjoint and cover each of $n_1, \ldots, n_j$.

Start with $A_0 = \varnothing$. Suppose A_{j-1} has been constructed. If $n_j \in A_{j-1} + B$, set $A_j = A_{j-1}$. Otherwise, for every $a \in A_{j-1}$ we have $n_j - a \notin B$, and therefore

$$C_j = \{n_j - a : a \in A_{j-1}\} \subset \mathbb{Z} \setminus B.$$

By the hypothesis, choose $b_j \in B$ such that $(C_j - b_j) \cap D = \varnothing$, and put

$$a_j = n_j - b_j, \qquad A_j = A_{j-1} \cup \{a_j\}.$$

Then $n_j = a_j + b_j \in a_j + B$. Moreover, for every old $a \in A_{j-1}$,

$$a_j - a = (n_j - a) - b_j \notin D.$$

Thus $(a_j + B) \cap (a + B) = \varnothing$ for every $a \in A_{j-1}$, because an intersection would imply $a_j - a \in B - B = D$. The induction continues.

Let

$$A = \bigcup_{j \geq 0} A_j.$$

Every integer is eventually covered, because it appears as some n_j. The translates $a + B$, $a \in A$, are pairwise disjoint by construction. Hence every integer has a unique representation as $a + b$ with $a \in A$ and $b \in B$. □

Proposition 1.8. *For $B = \{m^{13} : m \in \mathbb{Z}\}$, the hypothesis of Lemma 1.7 holds.*

Proof. Let $C \subset \mathbb{Z} \setminus B$ be finite. For a fixed $c \in C$, a choice $b = t^{13} \in B$ is bad for c precisely when

$$c - t^{13} \in D.$$

Since $D = B - B$ is symmetric, this is equivalent to

$$t^{13} - c \in D.$$

Thus the bad integers t with $|t| \leq T$ are exactly those counted by $S_c(T)$. By Proposition 1.6, for each fixed $c \in C$ this number is $o(T)$. Since C is finite, the union of all bad sets has size $o(T)$, whereas there are $2T + 1$ integers t with $|t| \leq T$. For all sufficiently large T, some integer t avoids every bad condition. Then $b = t^{13}$ satisfies

$$(C - b) \cap D = \varnothing.$$

This proves the proposition. □

Proof of Theorem 1.1. Proposition 1.8 verifies the hypothesis of Lemma 1.7 for $B = \{m^{13} : m \in \mathbb{Z}\}$. Hence there is a set $A \subset \mathbb{Z}$ such that every integer has a unique representation $n = a + b$ with $a \in A$ and $b \in B$. Since the map $m \mapsto m^{13}$ is injective on $\mathbb{Z}$, this is the same as a unique representation

$$n = a + m^{13}, \qquad a \in A, \quad m \in \mathbb{Z}.$$

The theorem follows. □

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